16

Troubleshooting Ethernet

Objectives

When you have completed study of this chapter, you will be able to identify, troubleshoot and fix problems such as:

- Thin and thick coax cable and connectors
- UTP cabling
- Incorrect media selection
- Jabber
- Too many nodes
- Excessive broadcasting
- Bad frames
- Faulty auto–negotiation
- 10/100 Mbps mismatch
- Full-/half-duplex mismatch
- Faulty hubs
- Switched networks
- Loading

16.1 Introduction

This section deals with addressing common faults on Ethernet networks. Ethernet encompasses layers 1 and 2, namely the physical and data Link layers of the OSI model. This is equivalent to the bottom layer (the network interface layer) in the ARPA model. This section will focus on those layers only, as well as on the actual medium over which the communication takes place.

16.2 Common problems and faults

Ethernet hardware is fairly simple and robust, and once a network has been commissioned, providing the cabling has been done professionally and certified, the network should be fairly trouble–free.

Most problems will be experienced at the commissioning phase, and could theoretically be attributed to the cabling, the LAN devices (such as hubs and switches), the network interface cards (NICs) or the protocol stack configuration on the hosts.

The wiring system should be installed and commissioned by a certified installer. The suppliers of high–speed Ethernet cabling systems, such as ITT, will in any case not guarantee their wiring if not installed by an installer certified by them. This effectively rules out wiring problems for new installations, although old installations could be suspect.

If the LAN devices such as hubs and switches are from reputable vendors, it is highly unlikely that they will malfunction in the beginning. Care should nevertheless be taken to ensure that intelligent (managed) hubs and switches are correctly set up.

This applies to NICs also. NICs rarely fail and nine times out of ten the problem lies with a faulty setup or incorrect driver installation or an incorrect configuration of the higher level protocols such as IP.

16.3 Tools of the trade

In addition to fundamental understanding of the technologies involved, spending sufficient time, employing a pair of eyes and patience, one can be successful in isolating Ethernet related problems with the help of the following tools:

16.3.1 Multimeters

A simple multimeter can be used to check for continuity and cable resistance, as will be explained in this section.

16.3.2 Handheld cable testers

There are many versions available in the market, ranging from simple devices that basically check for wiring continuity to sophisticated devices that comply with all the prerequisites for 1000BaseT wiring infrastructure tests. Testers are available from several vendors such as MicroTest, Fluke, and Scope.



Figure 16.1
Cable tester

16.3.3 Fiber optic cable testers

Fiber optic testers are simpler than UDP testers, since they basically only have to measure continuity and attenuation loss. Some UDP testers can be turned into fiber optic testers by purchasing an attachment that fits onto the existing tester. For more complex problems such as finding the location of a damaged section on a fiber optic cable, an alternative is to use a proper optical time domain reflectometer (OTDR) but these are expensive instruments and it is often cheaper to employ the services of a professional wire installer (with his own OTDR) if this is required.

16.3.4 Traffic generators

A traffic generator is a device that can generate a pre–programmed data pattern on the network. Although they are not used for fault finding strictly speaking, they can be used to predict network behavior due to increased traffic, for example, when planning network changes or upgrades. Traffic generators can be stand–alone devices or they can be integrated into hardware LAN analyzers such as the Hewlett Packard 3217.

16.3.5 RMON probes

An RMON (Remote MONitoring) probe is a device that can examine a network at a given point and keep track of captured information at a detailed level. The advantage of a RMON probe is that it can monitor a network at a remote location. The data captured by the RMON probe can then be uploaded and remotely displayed by the appropriate RMON management software. RMON probes and the associated management software are available from several vendors such as 3COM, Bay Networks and NetScout. It is also possible to create an RMON probe by running commercially available RMON software on a normal PC, although the data collection capability will not be as good as that of a dedicated RMON probe.

16.3.6 Handheld frame analyzers

Handheld frame analyzers are manufactured by several vendors, for up to Gigabit Ethernet speeds. These little devices can perform link testing, traffic statistics gathering etc. and can even break down frames by protocol type. The drawback of these testers is the small display and the lack of memory, which results in a lack of historical or logging functions on these devices.

Some frame testers are non–intrusive, i.e. they are clamp–style meters that simply clamp on to the wire and do not have to be attached to a hub port.



Figure 16.2
Psibernet Gigabit Ethernet probe

16.3.7 Software protocol analyzers

Software protocol analyzers are software packages running on PCs and using either a general purpose or a specialized NIC to capture frames from the network. The NIC is controlled by a so–called promiscuous driver, which enables it to capture all packets on the medium and not only those addressed to it in broadcast or unicast mode.

On the lower end of the scale, simple analyzers are available for download from the Internet as freeware or shareware. The free packages work well but rely heavily on the user for interpreting the captured information. Top of the range software products such as Network Associates' 'Sniffer' or WaveTek Wandel Goltemann's 'Domino' suite have sophisticated expert systems that can aid in the analysis of the captured software. Unfortunately, this comes at a price.

16.3.8 Hardware based protocol analyzers

Several manufacturers such as Hewlett Packard, Network Associates and WaveTek also supply hardware based protocol analyzers using their protocol analysis software running on a proprietary hardware infrastructure. This makes them very expensive but dramatically increases the power of the analyzer. For fast and gigabit Ethernet, this is probably the better approach.

16.4 Problems and solutions

16.4.1 Noise

If excessive noise is suspected on a coax or UTP cable, an oscilloscope can be connected between the signal conductor(s) and ground. This method will show up noise on the conductor, but will not necessarily give a true indication of the amount of power in the noise. A simple and cheap method to pick up noise on the wire is to connect a small loudspeaker between the conductor and ground. A small operational amplifier can be used as an input buffer, so as not to 'load' the wire under observation. The noise will be heard as an audible signal.

The quickest way to get rid of a noise problem, apart from using screened UTP (ScTP), is to change to a fiber–based instead of a wire–based network, for example, by using 100BaseFX instead of 100BaseTX.

Noise can to some extent be counteracted on a coax–based network by earthing the screen AT ONE END ONLY. Earthing it on both sides will create an earth loop. This is normally accomplished by means of an earthing chain or an earthing screw on one of the terminators. Care should also be taken not to allow contact between any of the other connectors on the segment and ground.

16.4.2 Thin coax problems

Incorrect cable type

The correct cable for thin Ethernet is RG58A/U or RG58C/U. This is a 5–mm diameter coaxial cable with 50–ohm characteristic impedance and a stranded center conductor. Incorrect cable used in a thin Ethernet system can cause reflections, resulting in CRC errors, and hence many retransmitted frames.

The characteristic impedance of coaxial cable is a function of the ratio between the center conductor diameter and the screen diameter. Hence other types of coax may closely resemble RG58, but may have different characteristic impedance.

Loose connectors

The BNC coaxial connectors used on RG58 should be of the correct diameter and should be properly crimped onto the cable. An incorrect size connector or a poor crimp could lead to intermittent contact problems, which are very hard to locate. Even worse is the 'Radio Shack' hobbyist type screw–on BNC connector that can be used to quickly make up a cable without the use of a crimping tool. These more often than not lead to very poor connections. A good test is to grip the cable in one hand, and the connector in another, and pull very hard. If the connector comes off, the connector mounting procedures need to be seriously reviewed.

Excessive number of connectors

The total length of a thin Ethernet segment is 185 m and the total number of stations on the segment should not exceed 30. However, each station involves a BNC T–piece plus two coax connectors and there could be additional BNC barrel connectors joining the cable. Although the resistance of each BNC connector is small, they are still finite and can add up. The total resistance of the segment (cable plus connectors) should not exceed 10 ohms otherwise problems can surface.

An easy method of checking the loop resistance (the resistance to the other end of the cable and back) is to remove the terminator on one end of the cable and measure the resistance between the connector body and the center contact. The total resistance equals the resistance of the cable plus connectors plus the terminator on the far side. This should be between 50 and 60 ohms. Anything more than this is indicative of a problem.

Overlong cable segments

The maximum length of a thin net segment is 185 m. This constraint is not imposed by collision domain considerations but rather by the attenuation characteristics of the cable. If it is suspected that the cable is too long, its length should be confirmed. Usually, the cable is within a cable trench and hence it cannot be visually measured. In this case, a time domain reflectometer (TDR) can be used to confirm its length.

Stub cables

For thin Ethernet (10Base2), the maximum distance between the bus and the transceiver electronics is 4 cm. In practice, this is taken up by the physical connector plus the PC board tracks leading to the transceiver, which means that there is no scope for a drop cable or 'stub' between the NIC and the bus. The BNC T–piece has to be mounted directly on to the NIC.

Users might occasionally get away with putting a short stub between the T–piece and the NIC but this invariably leads to problems in the long run.

Incorrect terminations

10Base2 is designed around 50 ohm coax and hence requires a 50 ohm terminator at each end. Without the terminators in place, there would be so many reflections from each end that the network would collapse. A slightly incorrect terminator is better than no terminator, yet may still create reflections of such magnitude that it affects the operation of the network.

A 93 ohm terminator looks no different than a 50 ohm terminator; therefore it should not be automatically assumed that a terminator is of the correct value.

If two 10Base2 segments are joined with a repeater, the internal termination on the repeater can be mistakenly left enabled. This leads to three terminators on the segment, creating reflections and hence affecting the network performance.

The easiest way to check for proper termination is by alternatively removing the terminators at each end, and measuring the resistance between connector body and center pin. In each case, the result should be 50 to 60 ohms. Alternatively, one of the T–pieces in the middle of the segment can be removed from its NIC and the resistance between the connector body and the center pin measured. The result should be the value of the two half cable segments (including terminators) in parallel, that is, 25 to 30 ohms.

Invisible insulation damage

If the internal insulation of coax is inadvertently damaged, for example, by placing a heavy point load on the cable, the outer cover could return to its original shape whilst leaving the internal dielectric deformed. This leads to a change of characteristic impedance at the damaged point resulting, in reflections. This, in turn, could lead to standing waves being formed on the cable.

An indication of this problem is when a work station experiences problems when attached to a specific point on a cable, yet functions normally when moved a few meters to either side. The only solution is to remove the offending section of the cable. It cannot be seen by the naked eye and the position of the damage has to be located with a TDR because of the nature of the damage. Alternatively, the whole cable segment has to be replaced.

Invisible cable break

This problem is similar to the previous one, with the difference that the conductor has been completely severed at a specific point. Despite the terminators at both ends of the cable, the cable break effectively creates two half segments, each with an un–terminated end, and hence nothing will work.

The only method to discover the location of the break is by using a TDR.

Thick coax problems

Thick coax (RG8), as used for 10Base5 or thick Ethernet, will basically exhibit the same problems as thin coax yet there are a few additional complications.

Loose connectors

10Base5 use N–type male screw on connectors on the cable. As with BNC connectors, incorrect procedures or a wrong sized crimping tool can cause sloppy joints. This can lead to intermittent problems that are difficult to locate.

Again, a good test is to grab hold of the connector and to try and rip it off the cable with brute force. If the connector comes off, it was not properly installed in the first place.

Dirty taps

The MAU transceiver is often installed on a thick coax by using a vampire tap, which necessitates pre-drilling into the cable in order to allow the center pin of the tap to contact the center conductor of the coax. The hole has to go through two layers of braided screen and two layers of foil. If the hole is not properly cleaned pieces of the foil and braid can remain and cause short circuits between the signal conductor and ground.

Open tap holes

When a transceiver is removed from a location on the cable, the abandoned hole should be sealed. If not, dirt or water could enter the hole and create problems in the long run.

Tight cable bends

The bend radius on a thick coax cable may not exceed 10 inches. If it does, the insulation can deform to such an extent that reflections are created leading to CRC errors. Excessive cable bends can be detected with a TDR.

Excessive loop resistance

The resistance of a cable segment may not exceed 5 ohms. As in the case of thin coax, the easiest way to do this is to remove a terminator at one end and measure the loop resistance. It should be in a range of 50 – 55 ohms.

UTP problems

The most commonly used tool for UTP troubleshooting is a cable meter or pair scanner. At the bottom end of the scale, a cable tester can be an inexpensive tool, only able to check for the presence of wire on the appropriate pins of a RJ–45 connector. High–end cable testers can also test for noise on the cable, cable length, and crosstalk (such as near end signal crosstalk or NEXT) at various frequencies. It can check the cable against CAT5/5e specifications and can download cable test reports to a PC for subsequent evaluation.

The following is a description of some wiring practices that can lead to problems.

Incorrect wire type (solid/stranded)

Patch cords must be made with stranded wire. Solid wire will eventually suffer from metal fatigue and crack right at the RJ–45 connector, leading to permanent or intermittent open connection/s. Some RJ–45 plugs, designed for stranded wire, will actually cut through the solid conductor during installation, leading to an immediate open connection. This can lead to CRC errors resulting in slow network performance, or can even disable a workstation permanently.

The permanently installed cable between hub and workstation, on the other hand, should not exceed 90 m and must be of the solid variety. Not only is stranded wire more expensive for this application, but the capacitance is higher, which may lead to a degradation of performance.

Incorrect wire system components

The performance of the wire link between a hub and a workstation is not only dependent on the grade of wire used, but also on the associated components such as patch panels, surface mount units (SMUs) and RJ–45 type connectors. A single substandard connector on a wire link is sufficient to degrade the performance of the entire link.

High quality fast and Gigabit Ethernet wiring systems use high–grade RJ–45 connectors that are visibly different from standard RJ–45 type connectors.

Incorrect cable type

Care must be taken to ensure that the existing UTP wiring is of the correct category for the type of Ethernet being used. For 10BaseT, Cat3 UTP is sufficient, while Fast Ethernet

(100BaseT) requires Cat5 and Gigabit Ethernet requires Cat5e or better. This applies to patch cords as well as the permanently installed ('infrastructure') wiring.

Most industrial Ethernet systems nowadays are 100BaseX based and hence use Cat5 wiring. For such applications, it might be prudent to install screened Cat5 wiring (ScTP) for better noise immunity. ScTP is available with a common foil screen around 4 pairs or with an individual foil screen around each pair.

A common mistake is to use telephone grade patch ('silver satin') cable for the connection between an RJ–45 wall socket (SMU) and the network interface card in a computer. Telephone patch cables use very thin wires that are untwisted, leading to high signal loss and large amounts of crosstalk. This will lead to signal errors causing retransmission of lost packets, which will eventually slow the network down.

'Straight' vs. crossover cable

A 10BaseT/100BaseTX patch cable consists of 4 wires (two pairs) with an RJ–45 connector at each end. The pins used for the TX and RX signals are 1, 2 and 3, 6. Although a typical patch cord has 8 wires (4 pairs), the 4 unused wires are nevertheless crimped into the connector for mechanical strength. In order to facilitate communication between computer and hub, the TX and RX ports on the hub are reversed, so that the TX on the computer and the RX on the hub are interconnected whilst the TX on the hub is connected to the RX on the hub. This requires a 'straight' interconnection cable with pin 1 wired to pin 1, pin 2 wired to pin 2 etc.

If the NICs on two computers are to be interconnected without the benefit of a hub, a normal straight cable cannot be used since it will connect TX to TX and RX to RX. For this purpose, a crossover cable has to be used in the same way as a 'null' modem cable. Crossover cables are normally color coded (for example, green or black) in order to differentiate them from straight cables.

A crossover cable can create problems when it looks like a normal straight cable and the unsuspecting person uses it to connect a NIC to a hub or a wall outlet. A quick way to identify a crossover cable is to hold the two RJ–45 connectors side by side and observe the colors of the 8 wires in the cable through the clear plastic of the connector body. The sequence of the colors should be the same for both connectors.

Hydra cables

Some 10BaseT hubs feature 50 pin connectors to conserve space on the hub. Alternatively, some building wire systems use 50 pin connectors on the wiring panels but the hub equipment has RJ–45 connectors. In both cases, hydra or octopus cable has to be used. This consists of a 50 pin connector connected to a length of 25 pair cable, which is then broken out as a set of 12 small cables, each with a RJ–45 connector. Depending on the vendor the 50–pin connector can be attached through locking clips, Velcro strips or screws. It does not always lock down properly, although at a glance it may seem so. This can cause a permanent or intermittent break of contact on some ports.

For 10BaseT systems, near end crosstalk (NEXT), which occurs when a signal is coupled from a transmitting wire pair to a receiving wire pair close to the transmitter, (where the signal is strongest) causes most problems. This is not a serious problem on a single pair cable, as only two pairs are used but on the 25 pair cable, with many signals in close proximity, this can create problems. It can be very difficult to troubleshoot since it will require test equipment that can transmit on all pairs simultaneously.

Excessive untwists

On Cat5 cable, crosstalk is minimized by twisting each cable pair. However, in order to attach a connector at the end the cable has to be untwisted slightly. Great care has to be taken since excessive untwists (more than 1 cm) is enough to create excessive crosstalk, which can lead to signal errors. This problem can be detected with a high quality cable tester.

Stubs

A stub cable is an abandoned telephone cable leading from a punch–down block to some other point. This does not create a problem for telephone systems but if the same Cat3 telephone cabling is used to support 10BaseT, then the stub cables may cause signal reflections that result in bit errors. Again, a high quality cable tester can only detect this problem.

Damaged RJ–45 connectors

On RJ–45 connectors without protective boots, the retaining clip can easily break off especially on cheaper connectors made of brittle plastic. The connector will still mate with the receptacle but will retract with the least amount of pull on the cable, thereby breaking contact. This problem can be checked by alternatively pushing and pulling on the connector and observing the LED on the hub, media coupler or NIC– wherever the suspect connector is inserted. Because of the mechanical deficiencies of RJ–45 connectors, they are not commonly used on industrial Ethernet systems.

T4 on 2 pairs

100BaseTX is a direct replacement for 10BaseT in that it uses the same 2 wire pairs and the same pin allocations. The only prerequisite is that the wiring must be Cat5.

100BaseT4, however, was developed for installations where all the wiring is Cat3, and cannot be replaced. It achieves its high speed over the inferior wire by using all 4 pairs instead of just 2. In the event of deploying 100BaseT4 on a Cat3 wiring infrastructure, a cable tester has to be used to ensure that in fact, all 4 pairs are available for each link and have acceptable crosstalk.

100BaseT4 required the development of a new physical layer technology, as opposed to 100BaseTX/FX that used existing FDDI technology. Therefore, it became commercially available only a year after 100BaseX and never gained real market acceptance. As a result, very few users will actually be faced with this problem.

Fiber optic problems

Since fiber does not suffer from noise, interference and crosstalk problems there are basically only two issues to contend with, namely, attenuation and continuity.

The simplest way of checking a link is to plug each end of the cable into a fiber hub, NIC or fiber optic transceiver. If the cable is all right, the LEDs at each end will light up. Another way of checking continuity is by using an inexpensive fiber optic cable tester consisting of a light source and a light meter to test the segment.

More sophisticated tests can be done with an optical time domain reflectometer (OTDR). OTDRs can not only measure losses across a fiber link, but can also determine the nature and location of the losses. Unfortunately, they are very expensive but most professional cable installers will own one.

10BaseFX and 100BaseFX use LED transmitters that are not harmful to the eyes, but Gigabit Ethernet uses laser devices that can damage the retina of the eye. It is therefore

dangerous to try and stare into the fiber (all systems are infrared and therefore invisible anyway).

Incorrect connector installation

Fiber optic connectors can propagate light even if the two connector ends are not touching each other. Eventually, the gap between fiber ends may be so far apart that the link stops working. It is therefore imperative to ensure that the connectors are properly latched.

Dirty cable ends

A speck of dust or some finger oil deposited by touching the connector end is sufficient to affect communication because of the small diameter of the fiber (8-62 microns) and the low light intensity. Dust caps must be left in place when the cable is not in use and a fiber optic cleaning pad must be used to remove dirt and oils from the connector point before installation to avoid this problem.

Component ageing

The amount of power that a fiber optic transmitter can radiate diminishes during the working lifetime of the transmitter. This is taken into account during the design of the link but in the case of a marginal design, the link could start failing intermittently towards the end of the design life of the equipment. A fiber optic power meter can be used to confirm the actual amount of loss across the link but an easy way to trouble shoot the link is to replace the transceivers at both ends of the link with new ones.

16.4.3 AUI problems

Excessive cable length

The maximum length of the AUI cable is 50 m assuming that it is a proper IEEE 802.3 cable. Some installations use lightweight office grade cables that are limited to 12 m in length. If these cables are too long, the excessive attenuation can lead to intermittent problems.

DIX latches

The DIX version of the 15 pin D–connector uses a sliding latch. Unfortunately, not all vendors adhere to the IEEE 802 specifications and some use lightweight latch hardware, which results in a connector that can very easily become unstuck. There are basically two solutions to the problem. The first solution is to use a lightweight (office grade) AUI cable, provided the distance would not be a problem. This places less stress on the connector. The second solution is to use a special plastic retainer such as the ‘ET Lock’ made specifically for this purpose.

SQE test

The signal quality error (SQE) test signal is used on all AUI based equipment to test the collision circuitry. This method is only used on the old 15 pin AUI based external

transceivers (MAUs) and sends a short signal burst (about 10 bit times in length) to the NIC just after each frame transmission. This tests both the collision detection circuitry and the signal paths. The SQE operation can be observed by means of an LED on the MAU.

The SQE signal is only sent from the transceiver to the NIC and not on to the network itself. It does not delay frame transmissions but occurs during the inter–frame gap and is not interpreted as a collision.

The SQE test signal must, however, be disabled if an external transceiver (MAU) is attached to a repeater hub. If this is not done, the hub will detect the SQE signal as a collision and will issue a jam signal. As this happens after each packet, it can seriously delay transmissions over the network. The problem is that it is not possible to detect this with a protocol analyzer.

16.4.4 NIC problems

Basic card diagnostics

The easiest way to check if a particular NIC is faulty is to replace it with another (working) NIC. Modern NICs for desktop PCs usually have auto–diagnostics included and these can be accessed, for example, from the device manager in MS Windows. Some cards can even participate in a card to card diagnostic. Provided there are two identical cards, one can be set up as an initiator and one as a responder. Since the two cards will communicate at the data link level, the packets exchanged will, to some extent, contribute to the network traffic but will not affect any other devices or protocols present on the network.

The drivers used for card auto–diagnostics will usually conflict with the NDIS and ODI drivers present on the host, and a message is usually generated, advising the user that the Windows drivers will be shut down, or that the user should re–boot in DOS.

With PCMCIA cards, there is an additional complication in that the card diagnostics will only run under DOS, but under DOS the IRQ (interrupt address) of the NIC typically defaults to 5, which happens to be the IRQ for the sound card. Therefore, the diagnostics will usually pass every test, but fail on the IRQ test. This result can then be ignored safely if the card passes the other diagnostics. If the card works, it works!

Incorrect media selection

Some cards support more than one medium, for example, 10Base2/10Base5, or 10Base5/10BaseT, or even all three. It may then happen that the card fails to operate since it fails to ‘see’ the attached medium.

It is imperative to know how the selection is done. Modern cards usually have an auto–detect function but this only takes place when the machine is booted up. It does NOT re–detect the medium if it is changed afterwards. Therefore, if the connection to a machine is changed from 10BaseT to 10Base2, for example, the machine has to be re–booted.

Some older cards need to have the medium set via a setup program, whilst even older cards have DIP switches on which the medium has to be selected.

Wire hogging

Older interface cards find it difficult to maintain the minimum 9.6 micro second inter-frame spacing (IFS) and as a result of this, nodes tend to return to and compete for access

to the bus in a random fashion. Modern interface cards are so fast that they can sustain the minimum 9.6 microsecond IFS rate. As a result, it becomes possible for a single card to gain repetitive sequential access to the bus in the face of slower competition and hence 'hogging' the bus.

With a protocol analyzer, this can be detected by displaying a chart of network utilization versus time and looking for broad spikes above 50 percent. The solution to this problem is to replace shared hubs with switched hubs and increase the bandwidth of the system by migrating from 10 – 100 megabits per second, for example.

Jabbers

A jabber is a faulty NIC that transmits continuously. NICs have a built–in jabber control that is supposed to detect a situation whereby the card transmits frames longer than the allowed 1518 bytes and shut the card down. However, if this does not happen, the defective card can bring the network down. This situation is indicated by a very high collision rate coupled with a very low or nonexistent data transfer rate. A protocol analyzer might not show any packets since the jabbering card is not transmitting any sensible data. The easiest way to detect the offending card is by removing the cables from the NICs or the hub one–by–one until the problem disappears in which case the offending card is located.

Faulty CSMA/CD mechanism

A card with a faulty CSMA/CD mechanism will create a large number of collisions since it transmits legitimate frames but does not wait for the bus to be quiet before transmitting. As in the previous case, the easiest way to detect this problem is to isolate the cards one by one until the culprit is detected.

Too many nodes

A problem with CSMA/CD networks is that the network efficiency decreases as the network traffic increases. Although Ethernet networks can theoretically utilize well over 90% of the available bandwidth, the access time of individual nodes increase dramatically as network loading increases. The problem is similar to that encountered on many urban roads during peak hours. During rush hours, the traffic approaches the design limit of the road. This does not mean that the road stops functioning. In fact, it carries a very large number of vehicles, but to get into the main traffic from a side road becomes problematic.

For office type applications, an average loading of around 30% is deemed acceptable while for industrial applications, 3% is considered maximum. Should the loading of the network be a problem, the network can be segmented using switches instead of shared hubs. In many applications, it will be found that the improvement created by changing from shared to switched hubs, is larger than the improvement to be gained by upgrading from 10 Mbps to Fast Ethernet.

Improper packet distribution

Improper packet distribution takes place when one or more nodes dominate most of the bandwidth. This can be monitored by using a protocol analyzer and checking the source address of individual packets. Another way of checking this easily is by using the NDG software Web Boy facility and checking the contribution of the most active transmitters.

Nodes like this are typically performing tasks such as video conferencing or database access, which require a large bandwidth. The solution to the problem is to give these

nodes separate switch connections or to group them together on a faster 100BaseT or 1000BaseT segment.

Excessive broadcasting

A broadcast packet is intended to reach all the nodes in the network and is sent to a MAC address of FF-FF-FF-FF-FF-FF. Unlike routers, bridges and switches forward broadcast packets throughout the network and therefore cannot contain the broadcast traffic. Too many simultaneous broadcast packets can degrade network performance.

In general, it is considered that if broadcast packets exceed 5% of the total traffic on the network, it would indicate a broadcast overload problem. Broadcasting is a particular problem with Netware servers and networks using NetBIOS/NetBEUI. Again, it is fairly easy to observe the amount of broadcast traffic using the WebBoy utility.

A broadcast overload problem can be addressed by adding routers, layer 3 switches or VLAN switches with broadcast filtering capabilities.

Bad packets

Bad packets can be caused by poor cabling infrastructure, defective NICs, external noise, or faulty devices such as hubs, devices or repeaters. The problem with bad packets is that they cannot be analyzed by software protocol analyzers.

Software protocol analyzers obtain packets that have already been successfully received by the NIC. That means they are one level removed from the actual medium on which the frames exist and hence cannot capture frames that are rejected by the NIC. The only solution to this problem is to use a software protocol analyzer that has a special custom NIC, capable of capturing information regarding packet deformities or by using a more expensive hardware protocol analyzer.

16.4.5 Faulty packets

Runts

Runt packets are shorter than the minimum 64 bytes and are typically created by a collision–taking place during the slot time.

As a solution, try to determine whether the frames are collisions or under–runs. If they are collisions, the problem can be addressed by segmentation through bridges and switches. If the frames are genuine under–runs, the packet has to be traced back to the generating node that is obviously faulty.

CRC errors

CRC errors occur when the CRC check at the receiving end does not match the CRC checksum calculated by the transmitter.

As a solution, trace the frame back to the transmitting node. The problem is either caused by excessive noise induced into the wire, corrupting some of the bits in the frames, or by a faulty CRC generator in the transmitting node.

Late collisions

Late collisions are typically caused when the network diameter exceeds the maximum permissible size. This problem can be eliminated by ensuring that the collision domains

are within specified values, i.e. 2500 meters for 10 Mbps Ethernet, 250 m for Fast Ethernet and 200 m for Gigabit Ethernet.

Check the network diameter as outlined above by physical inspection or by using a TDR. If that is found to be a problem, segment the network by using bridges or switches.

Misaligned frames

Misaligned frames are frames that get out of sync by a bit or two, due to excessive delays somewhere along the path or frames that have several bits appended after the CRC checksum.

As a solution, try and trace the signal back to its source. The problem could have been introduced anywhere along the path.

Faulty auto–negotiation

Auto–negotiation is specified for

- 10BaseT,
- 100BaseTX,
- 100BaseT2,
- 100BaseT4 and
- 1000BaseT.

It allows two stations on a link segment (a segment with only two devices on it) e.g. an NIC in a computer and a port on a switching hub to negotiate a speed (10/100/1000Mbps) and an operating mode (full/half duplex). If auto–negotiation is faulty or switched off on one device, the two devices might be set for different operating modes and as a result, they will not be able to communicate.

On the NIC side the solution might be to run the card diagnostics and to confirm that auto–negotiation is, in fact, enabled.

On the switch side, this depends on the diagnostics available for that particular switch. It might also be an idea to select another port, or to plug the cable into another switch.

10/100 Mbps mismatch

This issue is related to the previous one since auto–negotiation normally takes care of the speed issue.

Some system managers prefer to set the speeds on all NICs manually, for example, to 10 Mbps. If such an NIC is connected to a dual–speed switch port, the switch port will automatically sense the NIC speed and revert to 10 Mbps. If, however, the switch port is only capable of 100 Mbps, then the two devices will not be able to communicate.

This problem can only be resolved by knowing the speed (s) at which the devices are supposed to operate, and then by checking the settings via the setup software.

Full/half duplex mismatch

This problem is related to the previous two.

A 10BaseT device can only operate in half–duplex (CSMA/CD) whilst a 100BaseTX can operate in full duplex OR half–duplex.

If, for example, a 100BaseTX device is connected to a 10BaseT hub, its auto–negotiation circuitry will detect the absence of a similar facility on the hub. It will therefore know, by default, that it is ‘talking’ to 10BaseT and it will set its mode to half–duplex. If, however, the NIC has been set to operate in full duplex only, communications will be impossible.

16.4.6 Host related problems

Incorrect host setup

Ethernet only supplies the bottom layer of the DOD model. It is therefore able to convey data from one node to another by placing it in the data field of an Ethernet frame, but nothing more. The additional protocols to implement the protocol stack have to be installed above it, in order to make networked communications possible.

In industrial Ethernet networks, this will typically be the TCP/IP suite, implementing the remaining layers of the ARPA model as follows.

The second layer of the DOD model (the internet layer) is implemented with IP (as well as its associated protocols such as ARP and ICMP).

The next layer (the host–to host layer) is implemented with TCP and UDP.

The upper layer (the application layer) is implemented with the various application layer protocols such as FTP, Telnet etc. The host might also require a suitable application layer protocol to support its operating system in communicating with the operating system on other hosts, on Windows, that is NetBIOS by default.

As if this is not enough, each host needs a network 'client' in order to access resources on other hosts, and a network 'service' to allow other hosts to access its own resources in turn. The network client and network service on each host do not form part of the communications stack but reside above it and communicate with each other across the stack.

Finally, the driver software for the specific NIC needs to be installed, in order to create a binding ('link') between the lower layer software (firmware) on the NIC and the next layer software (for example, IP) on the host. The presence of the bindings can be observed, for example, on a Windows 95/98 host by clicking 'settings' –> 'control panel' –> 'networks'–>'configuration,' then selecting the appropriate NIC and clicking 'Properties' –> 'Bindings.'

Without these, regardless of the Ethernet NIC installed, networking is not possible.

Failure to log in

When booting a PC, the Windows dialogue will prompt the user to log on to the server, or to log on to his/her own machine. Failure to log in will not prevent Windows from completing its boot–up sequence but the network card will not be enabled. This is clearly visible as the LED's on the NIC and hub will not light up.

16.4.7 Hub related problems

Faulty individual port

A port on a hub may simply be 'dead.' Everybody else on the hub can 'see' each other, except the user on the suspect port. Closer inspection will show that the LED for that particular channel does not light up. The quickest way to verify this is to remove the UTP cable from the suspect hub port and plugging it into another port. If the LEDs light up on the alternative port, it means that the original port is not operational.

On managed hubs, the configuration of the hub has to be checked by using the hub's management software to verify that the particular port has not, in fact, been disabled by the network supervisor.

Faulty hub

This will be indicated by the fact that none of the LEDs on the hub are illuminated and that none of the users on that particular hub are able to access the network. The easiest to check this is by temporarily replacing the hub with a similar one and checking if the problem disappears.

Incorrect hub interconnection

If hubs are interconnected in a daisy chain fashion by means of interconnecting ports with a UTP cable, care must be taken to ensure that either a crossover cable is used or that the crossover/uplink port on one hub ONLY is used. Failure to comply with this precaution will prevent the interconnected hubs from communicating with each other although it will not damage any electronics.

A symptom of this problem will be that all users on either side of the faulty link will be able to see each other but nobody will be able to see anything across the faulty link. This problem can be rectified by ensuring that a proper crossover cable is being used or, if a straight cable is being used, that it is plugged into the crossover/uplink port on one hub only. On the other hub, it must be plugged into a normal port.

16.5 Troubleshooting switched networks

Troubleshooting in a shared network is fairly easy since all packets are visible everywhere in the segment and as a result, the protocol analysis software can run on any host within that segment. In a switched network, the situation changes radically since each switch port effectively resides in its own segment and packets transferred through the switch are not seen by ports for whom they are not intended.

In order to address the problem, many vendors have built traffic monitoring modules into their switches. These modules use either RMON or SNMP to built up statistics on each port and report switch statistics to switched management software.

Capturing the packets on a particular switched port is also a problem, since packets are not forwarded to all ports in a switch hence there is no place to plug in a LAN analyzer and view the packets.

One solution implemented by vendors is port liaising, also known as port mirroring or port spanning. The liaising has to be set up by the user and the switch copies the packets from the port under observation to a designated spare port. This allows the LAN user to plug in a LAN analyzer onto the spare port in order to observe the original port.

Another solution is to insert a shared hub in the segment under observation that is between the host and the switch port to which it was originally connected. The LAN analyzer can then be connected to the hub in order to observe the passing traffic.

16.6 Troubleshooting Fast Ethernet

The most diagnostic software is PC based and it uses a NIC with a promiscuous mode driver. This makes it easy to upgrade the system by simply adding a new NIC and driver. However, most PCs are not powerful enough to receive, store and analyze incoming data rates. It might therefore be necessary to rather consider the purchase of a dedicated hardware analyzer.

Most of the typical problems experienced with fast Ethernet, have already been discussed. These include a physical network diameter that is too large, the presence of Cat3 wiring in the system, trying to run 100BaseT4 on 2 pairs, mismatched 10BaseT/100BaseTX ports, and noise.

16.7 Troubleshooting Gigabit Ethernet

Although Gigabit Ethernet is very similar to its predecessors, the packets arrive so fast that they cannot be analyzed by normal means. A Gigabit Ethernet link is capable of transporting around 125 MB of data per second and few analyzers have the memory capability to handle this. Gigabit Ethernet analyzers such as those made by Hewlett Packard (LAN Internet Advisor), Network Associates (Gigabit Sniffer Pro) and WaveTech Wandel Goltemann (Domino Gigabit Analyzer) are highly specialized Gigabit Ethernet analyzers. They minimize storage requirements by filtering and analyzing capture packets in real time, looking for a problem. Unfortunately, they come at a price tag of around US$ 50 000.